SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 (REACH)

Revision date: 06-Jun-2019
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Version: 3.2

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SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name/designation:

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1.2. Relevant identified uses of the substance or mixture and uses advised against

Use of the substance/mixture:

Electrolyte for testing equipment used in electroplating.

Relevant identified uses:

	Sector of uses [SU]
SU 3:	Industrial uses: Uses of substances as such or in preparations at industrial sites
SU 22:	Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
	Product Categories [PC]
PC 21:	Laboratory chemicals
	Process categories [PROC]
PROC 15:	Use as laboratory reagent

1.3. Details of the supplier of the safety data sheet

Supplier (manufacturer/importer/only representative/downstream user/distributor):

Helmut Fischer GmbH
Industriestr.21
71069 Sindelfingen
Germany

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Telefax: +49 (0) 7031 303-710

E-mail: mail@helmut-fischer.de

Website: www.helmut-fischer.de

E-mail (competent person): anja.sehorz@helmut-fischer.de, lisa.brutsch@helmut-fischer.de

1.4. Emergency telephone number

Labor, +49 (0) 7031 303-416 (Only available during office hours.)

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]-:

Hazard classes and hazard categories	Hazard statements	Classification procedure
Skin corrosion/irritation *(Skin Corr. 1A)*	H314: Causes severe skin burns and eye damage.	Calculation method.
Serious eye damage/eye irritation *(Eye Dam. 1)*	H318: Causes serious eye damage.	Calculation method.

2.2. Label elements

Labelling according to Regulation (EC) No. 1272/2008 [CLP]

Hazard pictograms:

GHS05
Corrosion

Signal word: Danger

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Hazard components for labelling:
sodium hydroxide

hazard statements for health hazards	
H314	Causes severe skin burns and eye damage.

Supplemental Hazard information (EU): -

Precautionary statements Prevention	
P264	Wash hands thoroughly after handling.
P280	Wear protective gloves/protective clothing/eye protection/face protection.

Precautionary statements Response	
P301 + P330 + P331	IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER/doctor/....
P363	Wash contaminated clothing before reuse.

2.3. Other hazards

Adverse physicochemical effects:
No information available.

Adverse human health effects and symptoms:
No information available.

Adverse environmental effects:
No information available.

Other adverse effects:
No information available.

SECTION 3: Composition / information on ingredients

3.2. Mixtures

Description:
aqueous solution (sodium hydroxide)

Hazardous ingredients / Hazardous impurities / Stabilisers:

product identifiers	Substance name Classification according to Regulation (EC) No 1272/2008-[CLP]	Concen-tration
CAS No.: 7732-18-5 **EC No.:** 231-791-2	**water** The substance is classified as not hazardous according to regulation (EC) No 1272/2008 [CLP]-.	87 Wt %
CAS No.: 1310-73-2 **EC No.:** 215-185-5	**sodium hydroxide** Skin Corr. 1A **Danger** H314	13 Wt %

Full text of H- and EUH-phrases: see section 16.

SECTION 4: First aid measures

4.1. Description of first aid measures

General information:
In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible). In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).

Following inhalation:
Provide fresh air.
In case of respiratory tract irritation, consult a physician.

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In case of skin contact:
After contact with skin, wash immediately with plenty of water and soap.
Immediate medical treatment required because corrosive injuries that are not treated are hard to cure.
IF ON CLOTHING: Take off immediately all contaminated clothing.

After eye contact:
In case of contact with eyes flush immediately with plenty of flowing water for 10 to 15 minutes holding eyelids apart and consult an ophthalmologist.

After ingestion:
Rinse mouth thoroughly with water.
Do NOT induce vomiting.
Call a physician immediately.

4.2. Most important symptoms and effects, both acute and delayed
Causes severe skin burns and eye damage.

4.3. Indication of any immediate medical attention and special treatment needed
Treat symptomatically.

SECTION 5: Firefighting measures

5.1. Extinguishing media
Suitable extinguishing media:
Water Carbon dioxide (CO2) Foam alcohol resistant foam Dry extinguishing powder

Unsuitable extinguishing media:
Full water jet

5.2. Special hazards arising from the substance or mixture
Fire fighting water forms corrosive alkaline solutions - slip hazard!

5.3. Advice for firefighters
In case of fire: Wear self-contained breathing apparatus.

5.4. Additional information
The product itself does not burn. Co-ordinate fire-fighting measures to the fire surroundings.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. For non-emergency personnel
Personal precautions:
Use personal protection equipment.
See protective measures under point 7 and 8.

6.1.2. For emergency responders
No data available

6.2. Environmental precautions
Do not allow to enter into soil/subsoil. Do not allow to enter into surface water or drains.

6.3. Methods and material for containment and cleaning up
For cleaning up:
Absorb with liquid-binding material (e.g. sand, diatomaceous earth, acid- or universal binding agents).
Treat the recovered material as prescribed in the section on waste disposal.
Wash with plenty of water.

6.4. Reference to other sections
No data available

6.5. Additional information
See protective measures under point 7 and 8.

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SECTION 7: Handling and storage

7.1. Precautions for safe handling

Protective measures

Advices on safe handling:

When using do not eat, drink, smoke, sniff.
Wash hands before breaks and after work.

Fire prevent measures:
No special fire protection measures are necessary.

7.2. Conditions for safe storage, including any incompatibilities

Requirements for storage rooms and vessels:
Suitable container/equipment material: Material, alkali-resistant
Unsuitable container/equipment material: Aluminium

Hints on storage assembly:
Do not store together with: Strong acid

Storage class: 8B – Non-combustible corrosive substances

Further information on storage conditions:
Keep/Store only in original container.

7.3. Specific end use(s)

Recommendation:
Observe technical data sheet.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

No data available

8.2. Exposure controls

8.2.1. Appropriate engineering controls

All work processes must always be designed so that the following is excluded:
Eye contact
Skin contact .

8.2.2. Personal protection equipment

Eye/face protection:
Eye glasses with side protection
In case of increased risk, additionally Face protection umbrella

Skin protection:
Hand protection: For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.
Suitable material: NR (natural rubber, natural latex) CR (polychloroprene, chloroprene rubber) NBR (Nitrile rubber)
Thickness of the glove material: >0.4mm
Breakthrough time (maximum wearing time): >480 min

Respiratory protection:
Usually no personal respirative protection necessary.

Other protection measures:
Protective clothing: Suitable protective clothing: alkali-resistant
General health and safety measures: Wash hands before breaks and after work.
Provide eye shower and label its location conspicuously

8.2.3. Environmental exposure controls

No data available

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SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance

Physical state: Liquid
Colour: colourless
Odour: odourless
Odour threshold: not applicable

Safety relevant basis data

parameter		at °C	Method	Remark
pH	13			
Melting point	*not determined*		not determined	
Freezing point	*not determined*			
Initial boiling point and boiling range	*not determined*			
Decomposition temperature	*not determined*			
Flash point	*not determined*			
Evaporation rate	*not determined*			
Auto-ignition temperature	*not determined*			
Upper/lower flammability or explosive limits	*not determined*			
Vapour pressure	*not determined*			
Vapour density	*not determined*			
Density	*not determined*			
Bulk density	*not determined*			
Water solubility	completely miscible			
Partition coefficient: n-octanol/water	*not determined*			
Dynamic viscosity	*not determined*			
Kinematic viscosity	*not determined*			

9.2. Other information

No data available

SECTION 10: Stability and reactivity

10.1. Reactivity

Incompatible materials : Aluminium, Brass, Tin, zinc, glass

10.2. Chemical stability

The mixture is chemically stable under recommended conditions of storage, use and temperature.

10.3. Possibility of hazardous reactions

No hazardous reaction when handled and stored according to provisions.

10.4. Conditions to avoid

No information available.

10.5. Incompatible materials

Materials to avoid
Aluminium
Acid, concentrated

10.6. Hazardous decomposition products

No known hazardous decomposition products.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

CAS No.	Substance name	Toxicological information
1310-73-2	sodium hydroxide	**LD_{50} oral:** 2,000 mg/kg (Ratte)

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Acute oral toxicity:
There are no data available on the mixture itself.

Acute dermal toxicity:
There are no data available on the mixture itself.

Acute inhalation toxicity:
There are no data available on the mixture itself.

Skin corrosion/irritation:
Causes severe burns.

Serious eye damage/irritation:
Causes serious eye damage.

Respiratory or skin sensitisation:
not sensitising.

Germ cell mutagenicity:
There are no data available on the mixture itself.

Carcinogenicity:
There are no data available on the mixture itself.

Reproductive toxicity:
There are no data available on the mixture itself.

STOT-single exposure:
There are no data available on the mixture itself.

STOT-repeated exposure:
There are no data available on the mixture itself.

Aspiration hazard:
There are no data available on the mixture itself.

SECTION 12: Ecological information

12.1. Toxicity

Aquatic toxicity:
The product leads to changes in the pH value of the test system. The result refers to an unneutralised sample.

Additional ecotoxicological information:
The product is an alkali. Before discharge into sewage plants the product normally needs to be neutralised.

12.2. Persistence and degradability

CAS No.	Substance name	Biodegradation	Remark
1310-73-2	sodium hydroxide	not applicable	

Biodegradation:
The methods for determining the biological degradability are not applicable to inorganic substances.

12.3. Bioaccumulative potential

Accumulation / Evaluation:
No indication of bioaccumulation potential.

12.4. Mobility in soil

No adsoption in soil or sediment.

12.5. Results of PBT and vPvB assessment

CAS No.	Substance name	Results of PBT and vPvB assessment
1310-73-2	sodium hydroxide	The substance in the mixture does not meet the PBT/vPvB criteria according to REACH, annex XIII.

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, annex XIII.

12.6. Other adverse effects

No data available

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SECTION 13: Disposal considerations

13.1. Waste treatment methods

Consult the appropriate local waste disposal expert about waste disposal.
List of proposed waste codes/waste designations in accordance with EWC
This material and its container must be disposed of as hazardous waste.

13.1.1. Product/Packaging disposal

Waste codes/waste designations according to EWC/AVV

Waste code product:

16 05 06 *	laboratory chemicals, consisting of or containing hazardous substances, including mixtures of laboratory chemicals

*: Evidence for disposal must be provided.

Waste code packaging:

15 01 02	Plastic packaging

Waste treatment options

Appropriate disposal / Package:

Contaminated packages must be completely emptied and can be re-used following proper cleaning. This material and its container must be disposed of as hazardous waste.

SECTION 14: Transport information

Land transport (ADR/RID)	Inland waterway craft (ADN)	Sea transport (IMDG)	Air transport (ICAO-TI / IATA-DGR)
14.1. UN-No.			
UN 1824	UN 1824	UN 1824	UN 1824
14.2. UN proper shipping name			
SODIUM HYDROXIDE SOLUTION	SODIUM HYDROXIDE SOLUTION	SODIUM HYDROXIDE SOLUTION	SODIUM HYDROXIDE SOLUTION
14.3. Transport hazard class(es)			
 8	 8	 8	 8
14.4. Packing group			
II			
14.5. Environmental hazards			
No	No	No	No
14.6. Special precautions for user			
Special provisions: **Excepted Quantities:** **Hazard identification number (Kemler No.):** 80 **Classification code:** C5 **Remark:**	**Special provisions:** **Excepted Quantities:** **Classification code:** - **Remark:**	**Special provisions:** **Excepted Quantities:** **EmS-No.:** **Remark:**	**Special provisions:** **Excepted Quantities:** **Remark:**

14.7. Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

No data available

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. EU legislation

Restrictions on use:

Observe restrictions to employment for juvenils according to the 'juvenile work protection guideline' (94/33/EC).
Observe employment restrictions under the Maternity Protection Directive (92/85/EEC) for expectant or nursing mothers.

15.1.2. National regulations

[DE] National regulations

Water hazard class (WGK)

WGK:

keine Angabe

Source:

Self-classification

15.2. Chemical Safety Assessment

Chemical safety assessments for substances in this mixture were not carried out.

SECTION 16: Other information

16.1. Indication of changes

No data available

16.2. Abbreviations and acronyms

No data available

16.3. Key literature references and sources for data

No data available

16.4. Classification for mixtures and used evaluation method according to regulation (EC) No 1272/2008 [CLP]

Classification according to Regulation (EC) No 1272/2008 [CLP]-:

Hazard classes and hazard categories	Hazard statements	Classification procedure
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Serious eye damage/eye irritation *(Eye Dam. 1)*	H318: Causes serious eye damage.	Calculation method.

16.5. Relevant R-, H- and EUH-phrases (Number and full text)

Hazard statements	
H314	Causes severe skin burns and eye damage.

16.6. Training advice

No data available

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16.7. Additional information

The above information describes exclusively the safety requirements of the product and is based on our present-day knowledge. The information is intended to give you advice about the safe handling of the product named in this safety data sheet, for storage, processing, transport and disposal. The information cannot be transferred to other products. In the case of mixing the product with other products or in the case of processing, the information on this safety data sheet is not necessarily valid for the new made-up material.

This Safety Data Sheet was drawn up by
TÜV SÜD Industrie Service GmbH (see below),
based on data from the supplier, who is named in section 1
and who is responsible for this document.

TÜV SÜD Industrie Service GmbH

Department Environmental Service

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